

# Andrea Bonucci

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## Education

- Università Bocconi**, MA in Quantitative Finance and Risk Management Sep 2024 – Sep 2025
- Specialization in quantitative finance, with a strong focus on derivatives, trading strategies, and risk management in financial markets.
  - **Relevant Coursework:** Derivatives, Credit Risk Management, Fixed Income Securities, Market Risks, Stochastic Calculus for Finance, and Structured Products.
  - Practical focus on pricing and trading derivatives, managing portfolio risks, and applying stochastic models to financial decision-making.
- Vrije Universiteit Amsterdam**, Bachelor of Science in Mathematics Sep 2020 – Dec 2023
- **Thesis:** *\*Combinatorial Aspects of Surface Cluster Algebras and Applications to Frobenius' Conjecture\** (Grade: 9/10). This research involved applying algebraic concepts to modern mathematical theories, demonstrating a strong foundation in abstract mathematics.
  - **Honours:** Selected as one of three students to conduct independent research on advanced algebraic topics.
- Jumeirah Bacalaureate School**, International Bacalaureate Diploma Sep 2018 – June 2020
- Graduated with Distinction in Mathematics Higher Level.

## Professional Experience

- Software Developer**, Marxact, Hilversum, Netherlands Sep 2023 – Apr 2024
- Developed advanced software solutions in Python for GNSS receivers, focusing on tilt-compensation to improve positioning accuracy in challenging environments.
  - Implemented highly sophisticated mathematical algorithms, including the **Extended Kalman Filter (EKF)**, which was used to estimate precise positions in real-time, accounting for non-linearities in the system.
  - Collaborated with hardware engineers to integrate Python-based solutions, improving synchronization between software and hardware components.
- Mathematics Tutor**, Jumeirah Bacalaureate School, Dubai, UAE Sep 2018 – Jun 2020
- Provided personalized mathematics tutoring to high school students, helping them strengthen their understanding of complex topics such as calculus, algebra, geometry and statistics.

## University Projects

- Four Color Theorem**, Vrije Universiteit Amsterdam Jan 2023 – Feb 2023
- Designed an original algorithm for a computer-assisted proof of the Four Color Theorem.
- Frieze Patterns and Combinatorics**, Vrije Universiteit Amsterdam Jun 2022 – Jul 2022
- Conducted independent research on the geometry and combinatorial structure of frieze patterns.
- Criminology and Genetics Assessment**, Vrije Universiteit Amsterdam Jan 2022 – Feb 2022
- Applied stochastic processes and Hardy-Weinberg Equilibrium to evaluate DNA trace analysis accuracy in forensics.
- Epidemiology Study**, Vrije Universiteit Amsterdam Jun 2021 – Jul 2021
- Applied statistical methods to assess and model the spread and lethality of infectious diseases.

## Skills

- **Programming Languages:** Python, R, PHP, LaTeX, Mathematica, MATLAB, K (K3, Kdb+/Q).
- **Financial Markets Expertise:** US Markets, European Markets, Derivatives.
- **Quantitative Analysis:** Statistical Modeling, Data Analysis, Algorithm Optimization.
- **Mathematical Expertise:** Advanced Calculus, Rings and Fields, Complex Analysis, Galois Theory, Topology, Differential Geometry, Number Theory.